

1(a)(i)	<i>independent variable</i> type of sugar or glucose / monosaccharide and maltose / disaccharide ; <i>dependent variable</i> height / length, of gas ;	2
1(a)(ii)	to allow yeast / cells, to be evenly distributed ;	1
1(a)(iii)	any seven from: 1 same / stated, concentration of glucose and maltose solutions ; 2 same / stated, volume of each sugar solution ; 3 same / stated, volume of yeast suspension (for each tube) / equally divide yeast suspension between (each tube) ; 4 same / stated, volume of water in each (large) tube ; 5 method of maintaining a constant temperature ; 6 maintaining a constant pH and use a buffer ; 7 measure / record, height of gas / liquid, in set time or measure / record, time taken, to reach a set height of gas / liquid ; 8 use of a ruler / AW for measuring gas / liquid ; 9 control, using killed yeast / replace yeast with water ; 10 repeat at least twice / three replicates, and finding mean (for at least one sugar) ; 11 low risk ;	7

Question	Answer	Marks
1(b)(i)	<i>any one from:</i> divide the volume of gas collected in a stated time, by the (same) time ; 1 ÷ time to reach a stated volume of gas ; use a tangent to the line and divide volume by time / calculate gradient ;	1
1(b)(ii)	<i>any three from:</i> 1 rate of <u>respiration</u> is higher for glucose (than maltose) / ora ; 2 initial rate of reaction / respiration faster<u>er</u> ; 3 gas / CO₂, production is higher<u>er</u> with glucose / ora ; 4 maltose is broken down before it can be respired / glucose can be respired immediately ; 5 maltase / enzyme, needed to break down maltose ;	3
1(c)	<i>any three from:</i> 1 not subjective / ora ; 2 more accurate / ora ; 3 more precise / ora ; 4 can continuously record data (with data logger / computer) ; 5 meter can be used if colour blind / ora ; 6 meter does not pose a risk or hazard / ora ;	3